Section A: MCQs (10 Marks)

(Each question: 1 marks)

Q1. High Performance Computing (HPC) primarily focuses on: a) Increasing CPU clock speed ***b) Parallel execution of tasks*** c) Reducing storage size d) Improving graphics rendering	Remember
Q2. MPI is mainly used for: a) Shared memory programming ***b) Message passing between processes*** c) Job scheduling d) File system management	Remember
Q3. Which factor most strongly affects communication performance in HPC systems? a) Disk speed ***b) Network latency and bandwidth*** c) Cache size d) Clock frequency	Understand
Q4. Bandwidth refers to: a) Time to send a message ***b) Amount of data transferred per second***	Remember

c) Network delay before transfer starts d) CPU utilization	
Q5. In distributed memory systems, processes communicate using: a) Shared variables b) Locks and semaphores ***c) Messages*** d) Threads	Remember
Q6. SLURM is an example of a: a) Programming language b) Network protocol ***c) Job scheduler*** d) Compiler	Remember
Q7. A job queue in an HPC system is mainly used to: a) Store program output ***b) Manage job execution requests*** c) Balance CPU cache d) Monitor power usage	Understand
Q8. Which of the following is a collective MPI operation? a) MPI_Send b) MPI_Recv ***c) MPI_Bcast*** d) MPI_Test	Remember

Q9. Which speedup could be achieved according to Amdahl's Law for an infinite number of processors if 5% of a program is sequential and the remaining part is ideally parallel? a) Infinite speedup b) 5 c) 50 ***d) 20***	Apply
Q10. Which factor limits speedup according to Amdahl's Law? a) Number of processors ***b) Sequential portion of the program*** c) Network bandwidth d) Cache size	Understand

Section B: Short Answer (24 Marks)

(Each question: 3 marks)

Q11. In an HPC system, multiple processes running on different nodes communicate over a network to exchange data. Explain the difference between latency and bandwidth in this communication ?	Understand
Q12. In a shared HPC cluster, many users submit jobs that require CPUs, memory, and network resources. Why is a job scheduler necessary in such systems?	Understand
Q13. HPC applications can run on systems where processors either share memory or have separate local memories. Differentiate between shared memory and distributed memory architectures.	Understand
Q14. As the number of MPI processes increases, processes need to exchange data more frequently to stay synchronized. Explain why communication overhead generally increases with the number of processes.	Analyse
Q16. Two HPC applications run on the same cluster: • Application A performs frequent small message exchanges • Application B transfers large blocks of data infrequently Which application is more sensitive to network latency and why?	Analyse

Section C: Long Answer (10 Marks)

(Each question: 5 marks)

Q17. *An HPC application may show different performance behavior depending on its size when executed on a single node compared to execution across multiple nodes.* Evaluate whether executing an HPC application across multiple nodes is always beneficial compared to single-node execution. Justify your answer using architectural considerations.	Evaluate
Q18. *Two parallel applications perform the same amount of computation, but differ in how often they exchange data. One application communicates very frequently in small amounts, while the other communicates less frequently after doing more computation.* Evaluate how these communication patterns affect the scalability of applications in HPC systems. Explain your answer.	Evaluate